

DOP :
DOS :

NLP PRACTICAL MANUAL - Prof. D V Chandran

EXPT 7 : Implementing Part of Speech (POS) tagging with (NLTK/spacy/Stanford) in Python

AIM : WAP to implement Part of Speech (POS) tagging with NLTK/spacy/Stanford in Python.

THEORY :

- Part-of-speech tagging is the process of assigning a part- of-speech or other syntactic class marker to each word in a corpus.
- Because tags are generally also applied to punctuation, tagging requires that the punctuation marks (period, comma, etc) be separated off of the words. Thus tokenization is usually performed before, or as part of, the tagging process, separating commas, quotation marks, etc., from words, and disambiguating end-of-sentence punctuation (period, question mark, etc) from part-of-word punctuation (such as in abbreviations like e.g. and etc.) .
- The input to a tagging algorithm is a string of words and a specified tag-set. The output is a single best tag for each word.
- Tagging algorithm will have multiple tags for a single word and select only one best appropriate. Tagged output using the Penn Treebank tagset :
Book/VB that/DT flight/NN ./.
Does/VBZ that/DT flight/NN serve/VB dinner/NN ?/.
- Ambiguity : For example, book is ambiguous. That is, it has more than one possible usage and part-of-speech. It can be a verb (as in book that flight or to book the suspect) or a noun (hand me that book)
- Example : Does that flight serve dinner ? - that is determiner. I
thought that your flight was earlier. - that is complementizer.
- The problem of POS-tagging is to resolve these ambiguities, choosing the proper tag for the context.
- Because POS tagging is thus a disambiguation task.
- Most tagging algorithms fall into one of two classes: rule-based taggers and stochastic taggers.

POS tagging in NLTK/spacy/Stanford : (explore(the number of taggers))

ALGORITHM :

INPUT - OUTPUT :

INPUT : Book that flight.

OUTPUT : Book / VB that / DT flight / NN

INPUT : Hand me that book.

OUTPUT : Hand / VB me / NN that / DT book / NN

CONCLUSION :

PROGRAM AND OUTPUT PRINTOUT